

Tables

Table 1: Baseline characteristics of patient cohorts

	Males, 25–50 (n = 7,189,133)	Females, 25–50 (n = 9,679,018)	Males, 50–75 (n = 5,838,398)	Females, 50–75 (n = 6,862,253)
Target	Control: 7,145,962 (99.4%), Case: 43,171 (0.6%)	Control: 9,614,405 (99.3%), Case: 64,613 (0.7%)	Control: 5,816,266 (99.6%), Case: 22,132 (0.4%)	Control: 6,835,811 (99.6%), Case: 26,442 (0.4%)
Mean age at screening	39 years 8 months	39 years 2 months	59 years 9 months	59 years 8 months
Mean ER admissions	0.40	0.49	0.46	0.47
Mean DX codes	30.14	43.65	53.19	61.41
Mean RX codes	15.72	21.76	34.04	37.58
Mean PROC codes	39.24	58.68	66.21	75.92
Depression	Case: 17,240 (39.9%), Control: 439,197 (6.1%)	Case: 35,302 (54.6%), Control: 1,141,139 (11.9%)	Case: 9,814 (44.3%), Control: 348,736 (6.0%)	Case: 15,598 (59.0%), Control: 810,049 (11.9%)
TBI	Case: 1,008 (2.3%), Control: 38,681 (0.5%)	Case: 1,459 (2.3%), Control: 46,577 (0.5%)	Case: 713 (3.2%), Control: 33,632 (0.6%)	Case: 812 (3.1%), Control: 41,112 (0.6%)
PTSD	Case: 2,122 (4.9%), Control: 30,723 (0.4%)	Case: 5,669 (8.8%), Control: 75,410 (0.8%)	Case: 928 (4.2%), Control: 22,538 (0.4%)	Case: 1,569 (5.9%), Control: 31,496 (0.5%)
Substance Abuse	Case: 14,536 (33.7%), Control: 551,743 (7.7%)	Case: 17,474 (27.0%), Control: 526,828 (5.5%)	Case: 7,447 (33.6%), Control: 464,905 (8.0%)	Case: 7,369 (27.9%), Control: 409,794 (6.0%)
Anxiety	Case: 16,861 (39.1%), Control: 660,082 (9.2%)	Case: 33,061 (51.2%), Control: 1,414,191 (14.7%)	Case: 7,660 (34.6%), Control: 373,311 (6.4%)	Case: 12,787 (48.4%), Control: 807,085 (11.8%)
Schizophrenia	Case: 640 (1.5%), Control: 5,266 (0.1%)	Case: 681 (1.1%), Control: 4,848 (0.1%)	Case: 447 (2.0%), Control: 5,372 (0.1%)	Case: 570 (2.2%), Control: 6,797 (0.1%)
Disorders of adult personality and behavior	Case: 1,391 (3.2%), Control: 21,665 (0.3%)	Case: 2,629 (4.1%), Control: 25,437 (0.3%)	Case: 641 (2.9%), Control: 12,713 (0.2%)	Case: 984 (3.7%), Control: 13,156 (0.2%)
Insomnia	Case: 5,328 (12.3%), Control: 224,908 (3.1%)	Case: 10,261 (15.9%), Control: 408,854 (4.3%)	Case: 3,072 (13.9%), Control: 198,689 (3.4%)	Case: 4,771 (18.0%), Control: 359,023 (5.3%)
Mental and Behavioral Disorders	Case: 31,143 (72.1%), Control: 1,629,099 (22.8%)	Case: 51,973 (80.4%), Control: 2,812,472 (29.3%)	Case: 16,103 (72.8%), Control: 1,204,798 (20.7%)	Case: 21,614 (81.7%), Control: 1,847,372 (27.0%)

Table 2: ZeBRAperformance for the 1-month prediction horizon.

Sex	Age	Cohort	AUC	TPR	PPV	ACC	NPV	LR+	LR-
Males	25–50	All Patients	0.87 ± 0.01	0.57 ± 0.01	0.07 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	11.26 ± 0.13	0.45 ± 0.01
		PTSD	0.82 ± 0.02	0.38 ± 0.03	0.40 ± 0.02	0.90 ± 0.01	0.95 ± 0.01	7.77 ± 0.57	0.65 ± 0.03
		TBI	0.89 ± 0.02	0.58 ± 0.04	0.28 ± 0.01	0.94 ± 0.01	0.99 ± 0.01	11.60 ± 0.81	0.45 ± 0.04
		no MDD	0.81 ± 0.01	0.43 ± 0.01	0.03 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	8.57 ± 0.18	0.60 ± 0.01
	50–75	All Patients	0.89 ± 0.01	0.61 ± 0.01	0.05 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	12.13 ± 0.19	0.41 ± 0.01
		PTSD	0.83 ± 0.02	0.35 ± 0.04	0.26 ± 0.02	0.92 ± 0.01	0.97 ± 0.01	7.68 ± 0.86	0.68 ± 0.05
		TBI	0.89 ± 0.03	0.54 ± 0.05	0.21 ± 0.02	0.94 ± 0.01	0.99 ± 0.01	11.37 ± 1.04	0.49 ± 0.06
		no MDD	0.82 ± 0.01	0.46 ± 0.01	0.02 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	9.12 ± 0.28	0.57 ± 0.02
Females	25–50	All Patients	0.90 ± 0.01	0.61 ± 0.01	0.08 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	12.04 ± 0.11	0.42 ± 0.01
		PTSD	0.83 ± 0.01	0.37 ± 0.02	0.43 ± 0.01	0.90 ± 0.01	0.94 ± 0.01	7.49 ± 0.34	0.66 ± 0.02
		TBI	0.88 ± 0.02	0.52 ± 0.04	0.27 ± 0.01	0.94 ± 0.01	0.98 ± 0.01	10.47 ± 0.71	0.51 ± 0.04
		no MDD	0.84 ± 0.01	0.47 ± 0.01	0.03 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	9.28 ± 0.17	0.56 ± 0.01
	50–75	All Patients	0.91 ± 0.01	0.65 ± 0.01	0.05 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	12.89 ± 0.17	0.37 ± 0.01
		PTSD	0.84 ± 0.02	0.40 ± 0.03	0.34 ± 0.02	0.92 ± 0.01	0.96 ± 0.01	7.94 ± 0.67	0.64 ± 0.04
		TBI	0.90 ± 0.02	0.57 ± 0.05	0.21 ± 0.01	0.94 ± 0.01	0.99 ± 0.01	11.45 ± 0.97	0.45 ± 0.05
		no MDD	0.85 ± 0.01	0.50 ± 0.02	0.02 ± 0.01	0.95 ± 0.01	1.00 ± 0.01	9.92 ± 0.30	0.53 ± 0.02

Table 3: Calibration performance by sex and age stratum for 1 month horizon (see also Fig. 2a).

Sex	Age	Metric	Value*
Female	25–50	Brier score	0.006 (0.006, 0.006)
		Calibration intercept	-0.004 (-0.158, 0.100)
		Calibration slope	1.002 (0.962, 1.026)
Female	50–75	Brier score	0.003 (0.003, 0.003)
		Calibration intercept	-0.137 (-0.142, 0.289)
		Calibration slope	0.970 (0.959, 1.063)
Male	25–50	Brier score	0.006 (0.006, 0.006)
		Calibration intercept	-0.158 (-0.162, 0.110)
		Calibration slope	0.962 (0.952, 1.024)
Male	50–75	Brier score	0.003 (0.003, 0.003)
		Calibration intercept	-0.141 (-0.146, 0.120)
		Calibration slope	0.965 (0.950, 1.013)

* Rounded to three decimal places. Values in parentheses denote 95% confidence intervals obtained by bootstrap resampling of out-of-sample predictions.

Table 4: ICD-10 codes used to define the case cohort

Condition	Codes
Suicidal ideations	R45.851
Suicide attempts	T14.91 T14.91XA T14.91XD T14.91XS
Intentional self-harm	X71 X71.0XXA X71.0XXD X71.0XXS X71.1XXA X71.1XXD X71.1XXS X71.2XXA X71.2XXD X71.2XXS X71.3XXA X71.3XXD X71.3XXS X71.8XXA X71.8XXD X71.8XXS X71.9XXA X71.9XXD X71.9XXS X72 X72.XXXA X72.XXXD X72.XXXS X73 X73.0XXA X73.0XXD X73.0XXS X73.1XXA X73.1XXD X73.1XXS X73.2XXA X73.2XXD X73.2XXS X73.8XXA X73.8XXD X73.8XXS X73.9XXA X73.9XXD X73.9XXS X74 X74.01XA X74.01XD X74.01XS X74.02XA X74.02XD X74.02XS X74.09XA X74.09XD X74.09XS X74.8XXA X74.8XXD X74.8XXS X74.9XXA X74.9XXD X74.9XXS X75 X75.XXXA X75.XXXD X75.XXXS X76 X76.XXXA X76.XXXD X76.XXXS X77 X77.0XXA X77.0XXD X77.0XXS X77.1XXA X77.1XXD X77.1XXS X77.2XXA X77.2XXD X77.2XXS X77.3XXA X77.3XXD X77.3XXS X77.8XXA X77.8XXD X77.8XXS X77.9XXA X77.9XXD X77.9XXS X78 X78.0XXA X78.0XXD X78.0XXS X78.1XXA X78.1XXD X78.1XXS X78.2XXA X78.2XXD X78.2XXS X78.8XXA X78.8XXD X78.8XXS X78.9XXA X78.9XXD X78.9XXS X79 X79.XXXA X79.XXXD X79.XXXS X80 X80.XXXA X80.XXXD X80.XXXS X81 X81.0XXA X81.0XXD X81.0XXS X81.1XXA X81.1XXD X81.1XXS X81.8XXA X81.8XXD X81.8XXS X82 X82.0XXA X82.0XXD X82.0XXS X82.1XXA X82.1XXD X82.1XXS X82.2XXA X82.2XXD X82.2XXS X82.8XXA X82.8XXD X82.8XXS X83 X83.0XXA X83.0XXD X83.0XXS X83.1XXA X83.1XXD X83.1XXS X83.2XXA X83.2XXD X83.2XXS X83.8XXA X83.8XXD X83.8XXS

Table 5: Feature and Parameter Counts by Model

Model	# DX Features	# DX Train- able Parame- ters	# RX Features	# RX Train- able Parame- ters	# PROC Features	# PROC Train- able Parame- ters	# Total Train- able Parame- ters	# Total Features
Males, 25–50	19,496	134,556	1,940	116,938	21,225	141,514	393,008	42,662
Females, 25–50	20,578	201,655	2,020	180,761	22,081	225,930	608,346	44,680
Males, 50–75	19,042	94,322	1,992	86,716	22,246	101,299	282,337	43,281
Females, 50–75	19,553	118,395	2,035	108,893	22,561	130,015	357,303	44,150